Practice runs and test results

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use repeated trials to explain and support your design.

SUCCESS CRITERIA - I can...

1. Measure the actual course distance and run times.
2. Calculate average speed and describe variation.
3. Choose a final change supported by results from repeated trials.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “trial” and “mean” to explain your answer. Sentence starters are optional.

trial: one run or repetition of a test

mean: sum of the values divided by the number of values

evidence: observations or measurements that support an idea

01 Warm-up: bring back an earlier idea

A **50**-tooth driver turns a **350**-tooth gear. **Give ideal output**/input torque and speed ratios and say what you treated as true about energy losses.

Word help: torque = the turning effect of a force about an axis

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: mean time = sum of trial times/number of trials
average speed = distance/time.
Measure course length; keep release and timing consistent. Retain every trial and required build records.

Worked reasoning: Supplied **3 m** runs take **2.0**, **2.4**, **2.1 s**. Report all trials; their mean time is **2.17 s**. “Fastest run **2.0 s**” answers a different question from typical performance.

Your turn: Make a trial table and **mark both mean** and fastest time.

Sentence starter: I notice ... . This supports ... because ... .

Trial	Distance (m)	Time (s)	Speed (m/s)
1	____	____	____
2	____	____	____
3	____	____	____

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Complete a final labeled sketch or attach a photo with labels. **Name a feature** that supports straight motion and one that supports reliable electrical operation.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Marzano check - circle: ① ② ③

04 Apply the physics

A team’s supplied **4.0 m** run times are **5.0**, **5.2** and **4.8 s**. Fill the table and **calculate mean time**, the full time interval, its width and speed calculated from mean time. Retain every trial.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Trial	Distance (m)	Time (s)	Speed (m/s)
1	____	____	____
2	____	____	____
3	____	____	____

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

One run stops early after a loose connection. Should you erase it? **Explain how** to **record it fairly** and **decide whether** it belongs in a speed comparison.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Marzano check - circle: 1 2 3

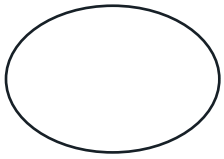
06 Review: connect an earlier idea

The same flux change through the same coil occurs in **1/4** of the time. **Compare induced-emf magnitude** and **name one thing** that must stay the same for that comparison.
Label the field or flux and **identify what changes**.

Word help: flux = a measure of magnetic field passing through a chosen area

Sentence starter: Both ... . They differ in ... because ... .

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: 1 2 3

07 Exit check

Explain a design choice in **three connected sentences**. **State your choice**, give the test results that support it, and **explain how those results support** your choice. Make clear whether you tested the cause by keeping other things the same or have only a possible explanation.

Sentence starter: My design includes ... . It does what is needed by ... .

Choice / claim	Actual evidence	Physics explanation
____	____	____
____	____	____

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

For **3 m** runs of **2.0** and **3.0 s**, compare mean of the two speeds with total distance divided by total time. **Explain why the results differ**.

Sentence starter: Both ... . They differ in ... because ... .

Trial	Distance (m)	Time (s)	Speed (m/s)
1	____	____	____
2	____	____	____
3	____	____	____

Marzano check - circle: 1 2 3